

A Fixed-Base Cognitive Potential Index

*A Bounded, Flynn-Adjusted Framework for Comparing Norm-Referenced
Cognitive Scores Across Time*

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Research status: proposed psychometric index and methodological framework. CPI is not a validated clinical IQ scale and should not be interpreted as a direct measure of innate intelligence.

Abstract

Intelligence quotients are norm-referenced scores: contemporary test norms are constructed so that the population mean remains approximately 100. Consequently, long-run secular changes in raw or age-standardized cognitive-test performance can be obscured when scores from different norming eras are compared directly. This paper proposes a Cognitive Potential Index (CPI), a fixed-base framework intended to preserve a bounded estimate of secular cognitive-performance change while retaining an individual's position relative to contemporary norms. The model separates observed secular score change, $R(t)$, from a transfer coefficient, $\lambda(t)$, representing the uncertain fraction of such change that can defensibly be interpreted as transferable latent cognitive improvement rather than test-specific, measurement, or environmental effects. The accumulated adjustment A is constrained to $0 \leq A \leq 100$ and is estimated from evidence rather than assumed to grow linearly. The generic model is $CPI = Q + A$, with A obtained from a bounded integral of λR . Existing meta-analytic evidence supports substantial historical Flynn gains but also shows heterogeneity by cognitive domain, declining gains in some recent cohorts, reversals in some populations, and weak or negative associations between secular gains and g -loadings. Accordingly, the paper treats A and λ as uncertain, context-dependent quantities and recommends hierarchical latent-variable and Bayesian estimation. A sensitivity illustration for 2026 demonstrates why the framework should report intervals and scenarios rather than a single universal conversion. The proposed CPI is therefore best understood as a research index for fixed-base temporal comparison, not as a replacement for standardized IQ testing.

Keywords: Flynn effect; intelligence testing; IQ norms; cognitive potential; secular trends; psychometrics; latent variable models; measurement invariance

1. Introduction

Modern intelligence tests report performance relative to a reference population rather than on an absolute physical scale. Re-norming is necessary for valid contemporary interpretation, but it complicates comparisons across distant norming eras. The Flynn effect - the secular increase in intelligence-test performance observed across many twentieth-century populations - makes this problem especially visible.

Two large meta-analyses provide useful empirical anchors. Trahan et al. (2014) estimated a mean Flynn effect of 2.31 IQ points per decade across 285 studies, while Pietschnig and Voracek (2015), using 271 independent samples and almost four million participants from 31 countries, reported domain-specific annual gains of approximately 0.41 points for fluid, 0.30 for spatial, 0.28 for full-scale, and 0.21 for crystallized performance. The latter analysis also found that gains had decreased in more recent decades.

A naive fixed-base index could simply accumulate a constant historical rate. That approach is attractive but untenable: secular changes differ by domain, country, age, test, and period, and some recent populations exhibit a reverse Flynn effect. This paper therefore develops a bounded and evidence-conditioned model in which secular gains are integrated over time but

discounted according to how much of the observed gain can be interpreted as transferable latent cognitive change.

2. Definitions and Scope

Let $Q_i(T,d)$ denote the conventional norm-referenced score of individual i at norming time T for cognitive domain d . Q is deliberately kept separate from the proposed historical adjustment: it retains its ordinary interpretation as standing relative to the relevant contemporary norm group.

$$CPI_i(T,d,x) = Q_i(T,d) + A(T,d,x) \quad (1)$$

Here x denotes additional contextual variables such as geography, age, socioeconomic environment, test family, and measurement conditions. A is the accumulated effective cognitive adjustment relative to a fixed historical baseline T_0 .

$$0 \leq A(T,d,x) \leq 100 \quad (2)$$

$$A(T_0,d,x) = 0 \quad (3)$$

The upper bound of 100 is a modeling constraint, not an empirical claim that human cognitive performance has a known biological ceiling exactly 100 points above the baseline. Its purpose is to prevent indefinite linear extrapolation and make the proposed index mathematically bounded.

3. From the Flynn Effect to an Accumulated Adjustment

Define $R(t,d,x)$ as the instantaneous secular change in fixed-base test performance. Positive values correspond to a Flynn-effect period, zero to a plateau, and negative values to a reverse-Flynn period.

$$R(t,d,x) = \partial G(t,d,x) / \partial t \quad (4)$$

If all observed score change were accumulated without qualification, the raw historical adjustment would be

$$G(T,d,x) = \int_{t_0}^T R(t,d,x) dt \quad (5)$$

However, G is not yet equivalent to cognitive potential. Secular test-score changes can reflect education, nutrition, familiarity with abstract problems, test-taking behavior, changing item properties, socioeconomic development, and other environmental or measurement processes. The model therefore introduces a transfer coefficient λ .

4. The Transfer Coefficient λ

Let $\lambda(t,d,x)$ represent the fraction of the local secular score change that the model attributes to transferable latent cognitive improvement for the purpose of CPI. It is bounded by construction:

$$0 \leq \lambda(t,d,x) \leq 1 \quad (6)$$

$\lambda = 1$ would treat the entire observed secular change as transferable cognitive gain; $\lambda = 0$ would treat none of it that way. Intermediate values represent partial transfer. Crucially, λ is not assumed to be universal or constant through time.

This caution is supported by the literature. Te Nijenhuis and van der Flier (2013) reported a true correlation of approximately -0.38 between subtest Flynn gains and g-loadings across their meta-analytic dataset. This does not provide a numerical estimate of λ ; rather, it warns against equating the entire Flynn effect with a rise in general intelligence. The method of correlated vectors and the interpretation of g-related secular change are themselves debated, so λ should be estimated using multiple forms of evidence rather than inferred from that correlation alone.

5. Generic CPI Model

The effective accumulated adjustment is defined by weighting each local secular change by its transfer coefficient and imposing the bounded interval $[0,100]$.

$$A(T,d,x) = \text{clip}_{(0,100)} \left[\int_{t_0}^T \lambda(t,d,x) R(t,d,x) dt \right] \quad (7)$$

where

$$\text{clip}_{(0,100)}(z) = \min(100, \max(0, z)). \quad (8)$$

Substituting Equation (7) into Equation (1) gives the generic master equation:

$$\text{CPI}_i(T,d,x) = Q_i(T,d) + \text{clip}_{(0,100)} \left[\int_{t_0}^T \lambda(t,d,x) R(t,d,x) dt \right] \quad (9)$$

For empirical use, both R and λ are uncertain. Let D denote the available evidence. The operational estimator is therefore posterior rather than deterministic:

$$\hat{A}(T,d,x) = E[A(T,d,x) | D] \quad (10)$$

$$\hat{\text{CPI}}_i = Q_i + \hat{A} \quad (11)$$

A credible or uncertainty interval should accompany $\hat{A}$. This prevents the model from hiding substantial uncertainty behind a single point estimate.

6. Empirical Constraints on $R(t)$

The historical literature rejects a single permanent slope. Trahan et al. (2014) estimated 2.31 points per decade overall, with a 95% confidence interval of 1.99 to 2.64. For modern Stanford-Binet and Wechsler comparisons, their estimate was 2.93 points per decade. Pietschnig and Voracek (2015) found stronger gains in fluid than crystallized performance and declining gains in recent decades.

More recent evidence reinforces the need for a time-varying R . Bratsberg and Rogeberg (2018) documented both the rise and reversal of the Flynn effect in Norwegian birth cohorts and showed that the pattern could be recovered within families, supporting environmental explanations. Dworak et al. (2023) reported declining scores in several domains in a large U.S. sample from 2006-2018 while performance on 3-D rotation increased. Thus, R can be positive, near zero, or negative depending on period and domain.

Evidence	Estimate / implication
Trahan et al. (2014)	Overall mean: 2.31 IQ points/decade; 95% CI 1.99-2.64.
Pietschnig & Voracek (2015)	Annual gains: fluid 0.41; spatial 0.30; full-scale 0.28; crystallized 0.21; gains decreased in recent decades.
te Nijenhuis & van der Flier (2013)	Meta-analytic correlation between Flynn gains and g-loadings ≈ -0.38 ; does not identify λ directly.
Bratsberg & Rogeberg (2018)	Rise and reversal observed within Norwegian families; supports environmental contribution.
Dworak et al. (2023)	Recent U.S. declines in several domains, with an increase in spatial rotation; modern trends are domain-specific.

7. Estimating λ and A

A direct universal estimate of λ is not currently identifiable from the cited literature. The proposed research program therefore treats λ as a latent, context-dependent parameter. One practical approach uses multi-group confirmatory factor analysis or item-response models to separate changes in latent ability from changes in intercepts, item difficulty, and residual structure.

$$X_j = \mu_j + \ell_j g + \varepsilon_j \quad (12)$$

For measurement-invariant indicators, a cross-cohort latent mean difference Δg can be estimated. A simple descriptive transfer ratio is then

$$\tilde{\lambda} = \text{clip}_{(0,1)} (\Delta g / \Delta IQ) \quad (13)$$

Equation (13) is only meaningful when Δg and ΔIQ are on compatible scales and measurement invariance is sufficiently supported. Otherwise, λ should be modeled jointly with item or subtest drift rather than calculated as a raw ratio.

A Bayesian implementation can impose the required bounds naturally. For example:

$$\lambda(t,d,x) \sim \text{Beta}(\alpha, \beta) \quad (14)$$

$$p(\lambda \mid D) \propto p(D \mid \lambda) p(\lambda). \quad (15)$$

A hierarchical model can allow partial pooling across domains, countries, tests, and periods, reducing overfitting while preserving genuine heterogeneity.

8. Illustrative 2026 Sensitivity Analysis

This section is illustrative rather than a validated conversion table. Taking the century-scale full-scale gain as an anchor gives approximately 29.1 accumulated raw points from 1909 to 2013 under the Pietschnig-Voracek average. Because post-2013 trends are heterogeneous, a provisional raw 2026 range of roughly 28-31 points is more defensible than extending 0.28 points per year mechanically.

Since λ is not identified by the current evidence, three sensitivity regimes can be displayed explicitly. These are modeling scenarios, not published estimates of the true transfer fraction.

Scenario	λ	If $G(2026)=30$, $A=\lambda G$	CPI for $Q=130$
Strict transfer	0.15	4.5	134.5
Balanced working scenario	0.35	10.5	140.5
High-transfer scenario	0.60	18.0	148.0

The example shows why a score such as “CPI = 160” should not be asserted merely by adding the full historical Flynn gain to a contemporary IQ score. Until λ is empirically estimated, the appropriate output is a sensitivity interval or posterior distribution.

9. Interpretation of CPI

CPI is designed to answer a narrow counterfactual comparison question: how might a contemporary norm-referenced score be represented on a fixed historical base after applying an explicitly bounded and evidence-weighted estimate of secular cognitive change? It does not claim that a CPI of 140 is psychometrically identical to a clinical IQ of 140, nor that people from later cohorts possess proportionally greater innate intelligence.

For this reason, reporting the two-component representation alongside the scalar index is recommended:

$$\text{CPI}^* = (Q, A) \quad (16)$$

For example, (130, 10.5) transparently communicates a contemporary norm-referenced score of 130 and a modeled historical adjustment of 10.5. The scalar $\text{CPI} = 140.5$ can then be used for fixed-base comparisons without concealing how it was constructed.

10. Testable Predictions and Validation

The framework yields several empirical requirements. First, estimated R should reproduce known cross-norm Flynn effects when integrated over the corresponding periods. Second, domain-specific models should outperform a universal-rate model when fluid, spatial, crystallized, and full-scale data are jointly evaluated. Third, a latent-variable estimate of λ should improve out-of-sample prediction of cross-cohort performance on novel, measurement-

invariant tasks if λ captures transferable ability rather than mere test familiarity. Fourth, the model should permit λ near zero or negative R where the data demand it; otherwise it would be unfalsifiable.

Validation should therefore use held-out cohorts, preregistered model comparisons, measurement-invariance testing, and posterior predictive checks. CPI should not be used for clinical, educational, employment, or legal decisions unless and until such validation exists.

11. Limitations

- IQ scores are norm-referenced constructs, not physical units; adding a modeled historical adjustment creates an index rather than an absolute intelligence scale.
- The choice of historical baseline affects interpretation. The year 1905 is conceptually associated with early modern intelligence testing, while large meta-analytic evidence begins later; empirical implementations should state the exact baseline and calibration procedure.
- The bound $A \leq 100$ is a design constraint and requires sensitivity analysis. Alternative saturating transformations may be preferable if future evidence supports them.
- λ is presently only partially identified. Correlations between Flynn gains and g-loadings do not directly equal the fraction of gains representing latent cognitive improvement.
- Measurement invariance across cohorts cannot be assumed. Item content, schooling, test familiarity, sampling, and demographic composition can all alter observed trends.
- Global averages can conceal substantial geographic and socioeconomic heterogeneity.
- Online IQ tests may not have transparent norming samples or contemporary representative norms and therefore should not be treated as equivalent to standardized clinical batteries.

12. Research Program

A rigorous empirical CPI project would assemble repeated-norm datasets from major intelligence batteries and large cohort studies, harmonize subtests into cognitive domains, test longitudinal and cross-cohort measurement invariance, estimate $R(t,d,x)$ using hierarchical smoothers, estimate λ through latent-variable changes on invariant indicators, and propagate all parameter uncertainty into A and CPI. The final model should be evaluated against simpler alternatives, including a constant Flynn-rate model and a no-adjustment model.

A useful extension would replace hard clipping with a smooth saturating transformation. For example, if $B(T)$ is the unconstrained effective accumulation, one may define

$$A(T) = 100 [1 - \exp(-\max(0, B(T))/100)] \quad (17)$$

which approaches 100 asymptotically rather than hitting the boundary abruptly. Whether clipping or smooth saturation is preferable is an empirical and modeling question, not something that should be fixed solely for convenience.

13. Conclusion

This paper proposes a bounded fixed-base Cognitive Potential Index for temporal comparison of norm-referenced cognitive scores. The model retains conventional IQ Q while adding an

explicitly modeled adjustment A. Unlike a simple linear Flynn correction, A is generated by integrating a time-, domain-, and context-dependent secular rate R after weighting it by an uncertain transfer coefficient λ . The resulting master equation is

$$CPI_i(T,d,x) = Q_i(T,d) + \text{clip}_{(0,100)} \left[\int_{t_0}^T \lambda(t,d,x) R(t,d,x) dt \right] \quad (18)$$

The principal contribution is therefore conceptual and methodological: it separates norm-relative standing, observed secular score change, and the stronger claim of transferable cognitive improvement. Existing Flynn-effect evidence supports substantial historical score gains but does not justify treating every gained IQ point as an equivalent increase in general cognitive potential. CPI consequently remains a proposed research index whose usefulness depends on future empirical estimation of λ , careful treatment of measurement invariance, and external validation.

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Appendix A. Symbol Table

Symbol	Definition
CPI_i	Cognitive Potential Index for individual i.
Q_i	Conventional norm-referenced cognitive/IQ score.
T	Norming or comparison time.
T_0	Fixed historical baseline.
d	Cognitive domain.
x	Vector of contextual variables.
$R(t,d,x)$	Instantaneous secular change in fixed-base

	test performance.
$\lambda(t,d,x)$	Transfer coefficient, constrained to [0,1].
$G(T,d,x)$	Raw accumulated secular score change.
$A(T,d,x)$	Bounded effective accumulated cognitive adjustment.
D	Empirical evidence used for estimation.
$\hat{A}$	Posterior mean or other chosen estimator of A.